AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter
Monitoring & Health Effects

Issue 2026-10-15 | Entry window 15 October 2026 (full day) | 25 records screened in scope

25 RECORDS IN SCOPE (85 EXCLUDED)

8 OF 10 SUBTOPIC BANDS POPULATED

4 EFFECT ESTIMATES WITH A CI

5 TIER-A RECORDS

5 RECORDS CARRIED ON METADATA ALONE

Signal of the day

AERONET's night-time lunar AOD now matches its daytime solar AOD to within ±0.01 at 440 nm across 492 sites — the column reference for AOD–$PM_{2.5}$ fusion no longer stops at sunset.

Model-T CIMEL radiometers have recorded direct-beam lunar irradiance at six wavelengths (440–1640 nm) since 2014. With an updated empirical correction to the ROLO lunar irradiance model, day–night AOD differences at every long-record site are typically **within ±0.01** at 440 nm and as good or better at longer wavelengths; in 3 h dusk/dawn transition windows under low, stable AOD, AOD, Ångström exponent, column water vapour and SDA fine/coarse AOD differ negligibly. **Where it is weak:** consistency is demonstrated only where AOD is low and stable — smoke and dust nights, the case that matters for exposure, are not tested, and lunar phase leaves recurring multi-day gaps. **Why it matters here:** satellite AOD is daytime-only, so AOD-driven $PM_{2.5}$ surfaces interpolate the night; a validated night column is a direct check on how ground LCS networks and models handle the nocturnal boundary layer.

What changed today

- **Low-cost monitors in a transfer-learning exposure model.** Lima 2010–2023: reference and low-cost monitors, MAIAC AOD and WRF-Chem fused by an instance-weighted random forest that borrows US training data by geo-environmental similarity. CV R^2 **0.76**; mountain-district mean $PM_{2.5}$ fell from 34.2 to 29.2 $\mu g\,m^{-3}$ — a correction that the reported CV, run on the same biased sites, cannot verify.
- **Datacenter power as a $PM_{2.5}$ exposure.** Memphis preprint (Dominici group): on-site turbines concentrate attributable deaths within 50 km (82 %), grid supply scatters them >250 km (94 %); at real/permitted capacity, **11.35** (6.48–17.29) vs **8.10** (4.60–12.38) deaths/yr. A near-source burden that a local network could test directly.
- **Two well-powered nulls.** MESA (n=5,072): no pollutant associated with suPAR, $PM_{2.5}$ CI excluding ≥2 % per 5 $\mu g\,m^{-3}$. UK Biobank women (n=231,525): the $PM_{2.5}$–NO_2 association with fatty-liver index vanishes after area deprivation (OR **0.999**, 0.993–1.004) — with the caveat that deprivation may sit on the pathway.
- **Constituents and windows.** Guangdong CHD registry (12,680 cases): per-IQR constituent increments raised odds 9–16 %, strongest preconception; conotruncal ORs 1.14–1.36; sulfate and BC lead early, BC and nitrate later. CIs absent from the abstract.
- **Non-exhaust metrology with explicit drift.** DLR tire-wear rig: chamber temperature raised CPC response 3.6-fold (1.8–7.2), inflation pressure cut OPS number 18 % (12–24 %), and the reference-condition CPC signal doubled over the campaign — drift monitoring and randomised blocks are prerequisites.

Window continuity

The 14 October issue closed with `last_entry_date` at 14 October. This issue collects **15 October in full**, continuous with it. 16 October is not opened: this run began at 11:23 CDT on 16 October, before 22:00. The **W42 weekly** (11–17 October) is due with the Saturday 17 October issue.

Corpus analytics

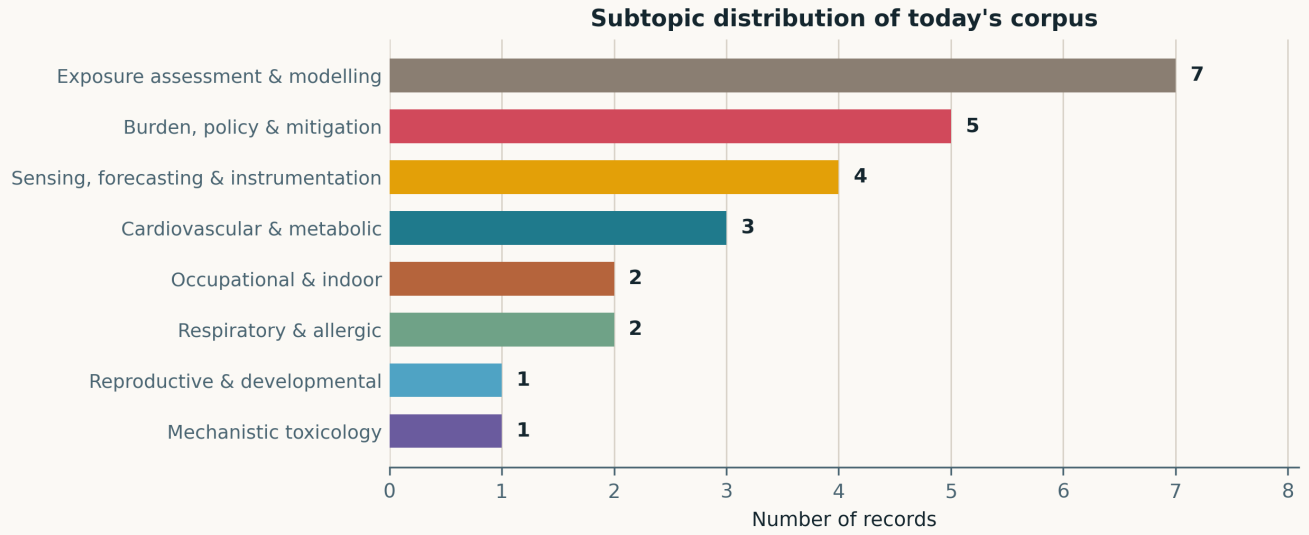


Fig. Subtopic assignment of the 25 in-scope records across eight bands. Exposure assessment and modelling leads (7), burden and policy follows (5); sensing (4) carries the AERONET lunar-AOD tier-A record. Neuro and other-clinical bands are empty.

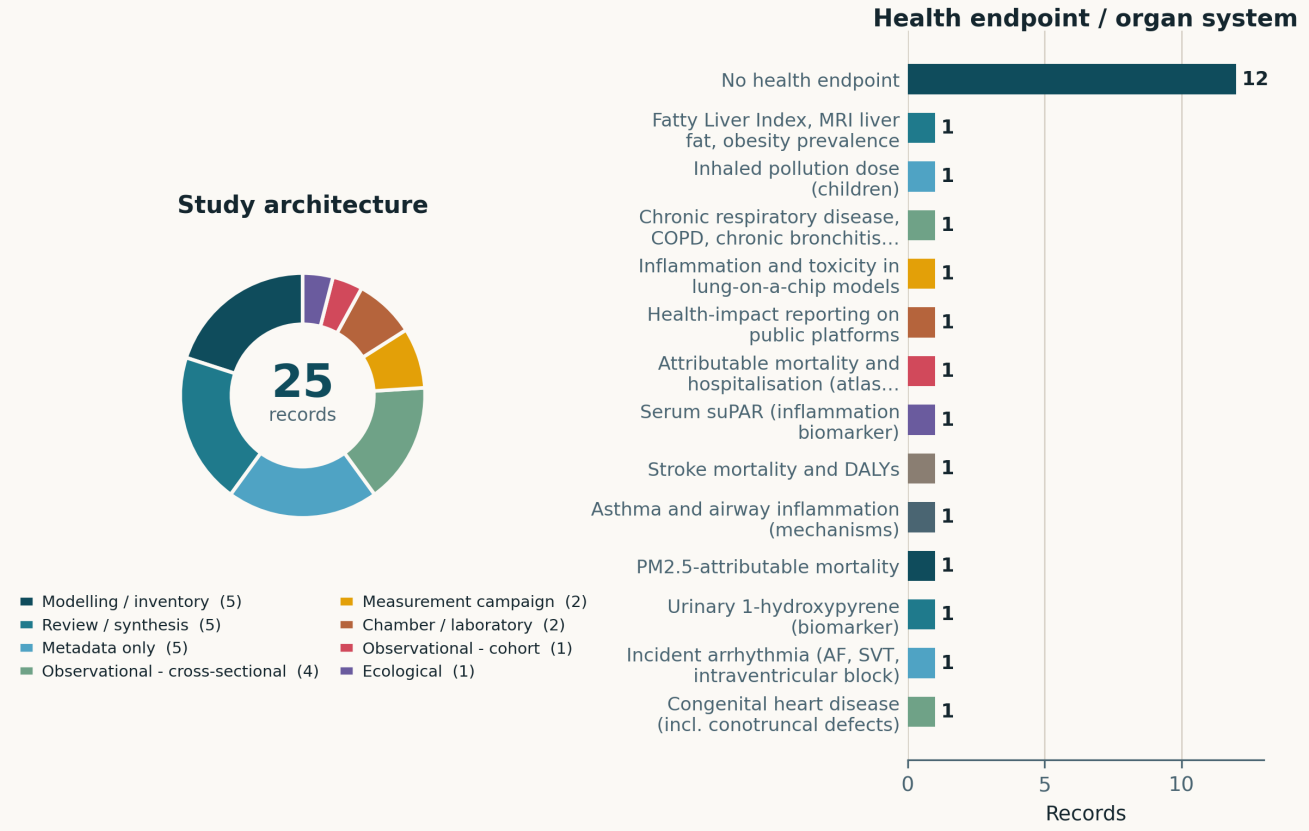


Fig. Left — architecture: modelling/inventory and review each 5, metadata-only 5 (20 %, all Elsevier or Springer Nature deposits without abstracts at harvest). Right — health endpoints: CHD, arrhythmia, chronic respiratory disease, stroke burden, fatty liver, a biomarker null (suPAR).

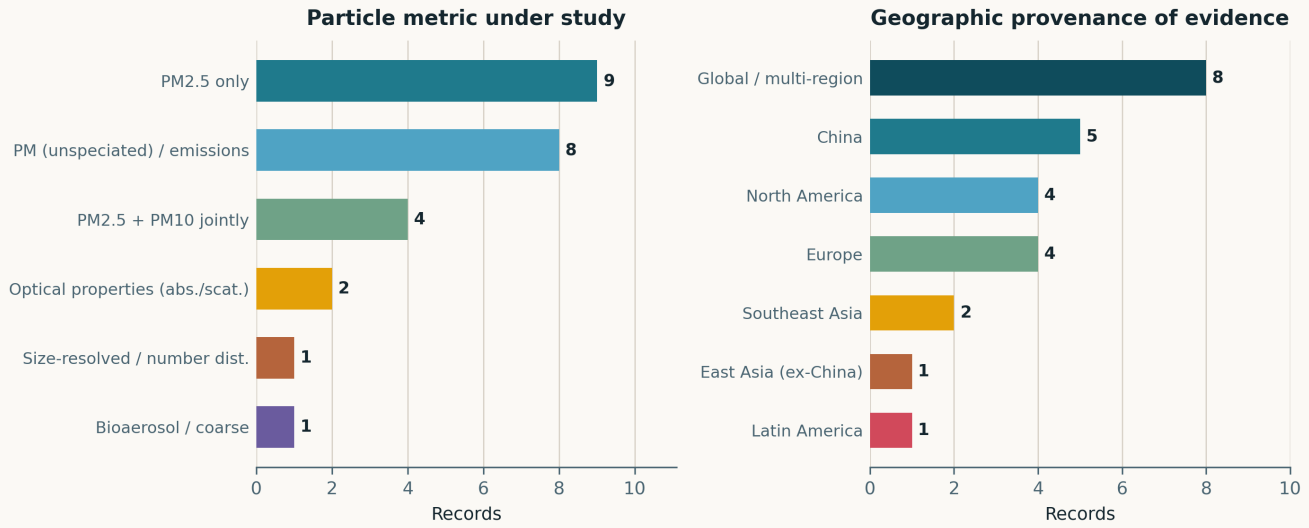


Fig. Left — particle metric: $PM_{2.5}$ dominates; optical (AOD) and size-resolved number (tire-wear CPC/OPS) metrics appear on the sensing side. Right — geography: China (5) and North America (4) lead; Peru, Indonesia/Malaysia and Taiwan enter.

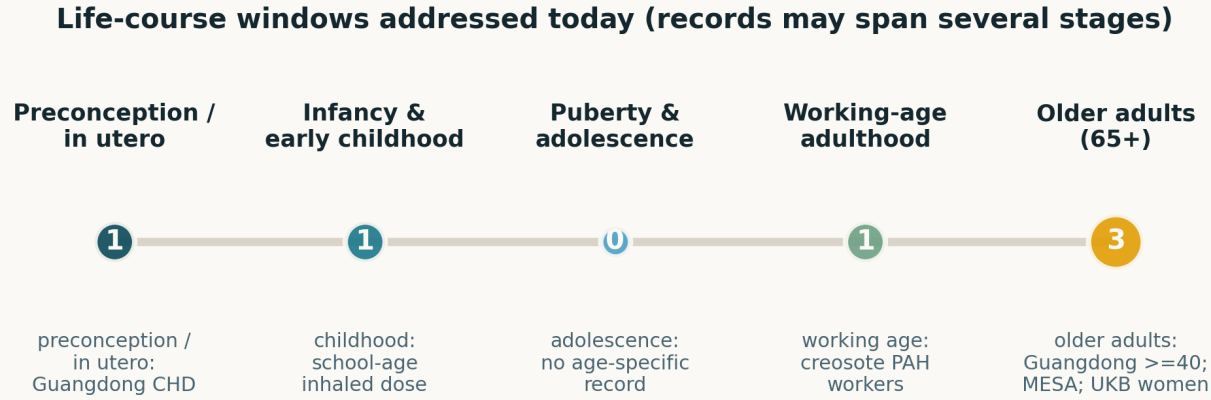


Fig. Life-course windows: preconception/in utero (Guangdong CHD), school-age children (inhaled dose), working age (creosote workers), and three older-adult samples (Guangdong ≥40, MESA, UK Biobank women). No adolescent record.

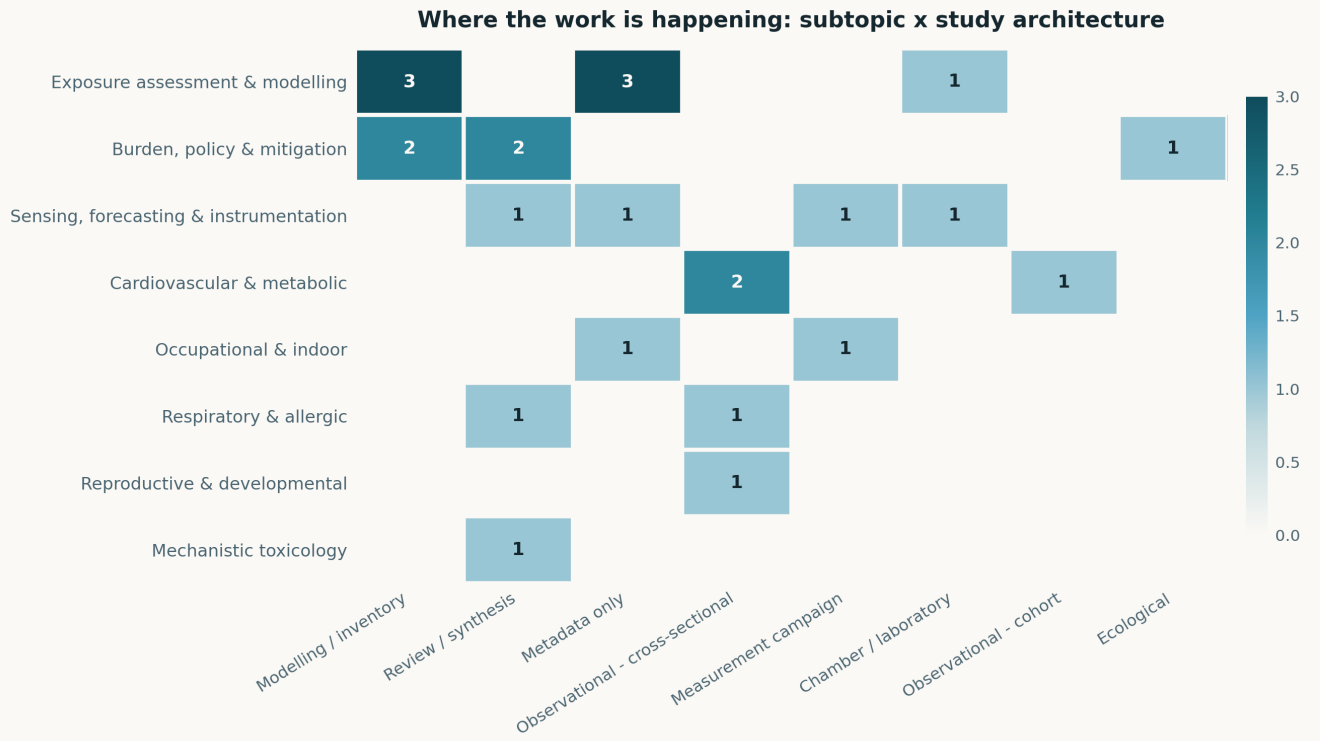


Fig. Subtopic × study architecture. Review/synthesis spreads across burden, respiratory, sensing and mechanism; the exposure row mixes modelling, laboratory and metadata-only records.

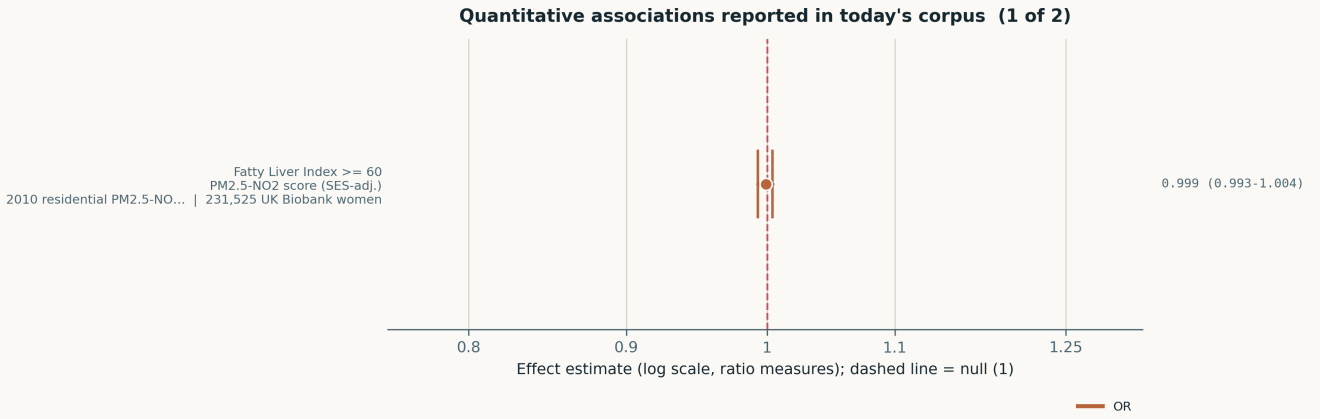


Fig. Ratio scale: one estimate — the UK Biobank fatty-liver OR per unit of a 2010 $PM_{2.5}$–NO_2 score after area-deprivation adjustment, a null by construction of the adjustment set, not per unit concentration.

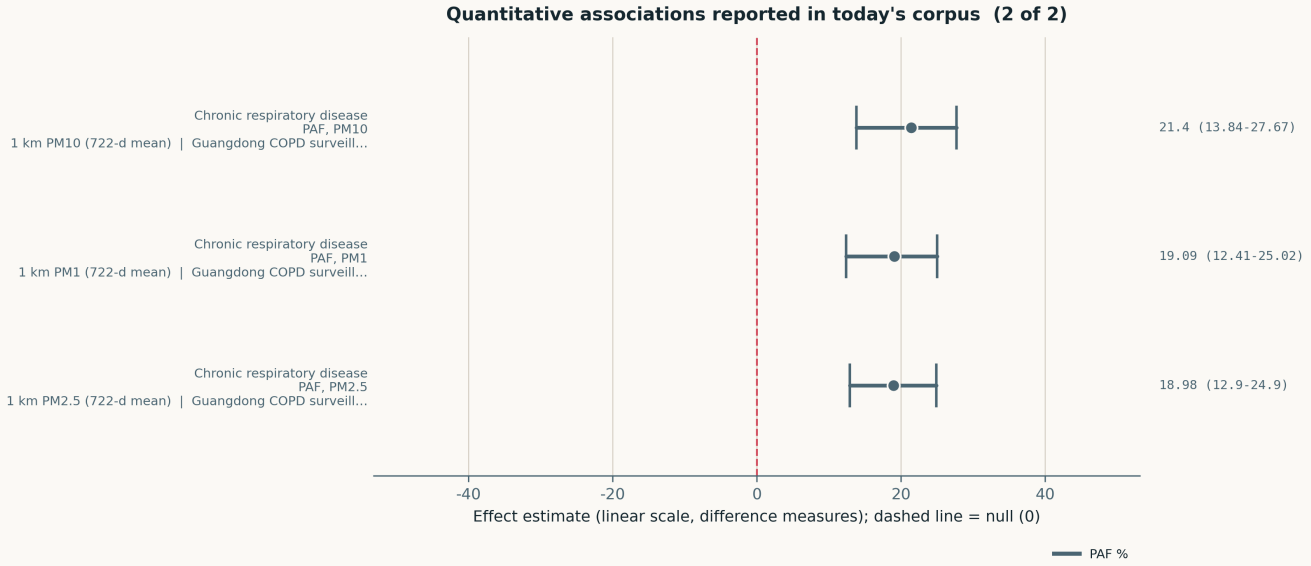


Fig. Difference scale: population-attributable fractions (%) of chronic respiratory disease for PM_1, $PM_{2.5}$ and PM_{10} from one cross-sectional Guangdong sample. All three come from the same 1 km satellite-ML product and are near-collinear — one estimate drawn three times, causal only by assumption.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Cardiovascular & metabolic — 3 records

UKB arrhythmia NMR metabolome mediation 2026 Ecotoxicol Environ Saf

Large-scale plasma metabolic profiling identifies signatures and biomarkers associated with air pollution and arrhythmias: A prospective cohort study from UK Biobank

449,068 UK Biobank participants free of arrhythmia, median follow-up 16.3 y, 47,381 incident cases. All four pollutants were associated with arrhythmia, HR 1.015-1.257 per 10 $\mu g\,m^{-3}$ (range across pollutants; CIs not in the abstract). An elastic-net signature from 170 NMR metabolites mediated 11.2% of the $PM_{2.5}$ association (PM_{10} 5.2%, NO_2 7.4%, NOx 9.8%); lipid, ketone-body and amino-acid pathways recurred, ketone bodies most for intraventricular block. Limitation: single-baseline 2010 LUR exposure and single-time metabolomics make mediation cross-sectional in disguise; ketone bodies track fasting and diabetes status. Relevance: low-moderate for sensing - another UKB LUR result whose exposure contrast is too smooth to need dense monitoring.

doi: 10.1016/j.ecoenv.2026.120919

MESA long-term air pollution suPAR null 2026 Environ Epidemiol

Long-term ambient air pollution is not associated with circulating soluble urokinase plasminogen activator receptor

5,072 MESA participants (2000-2002 enrolment), ELISA suPAR, multivariable linear regression. No pollutant - $PM_{2.5}$, NO_2, NOx, O_3 or light-absorbing carbon - was associated with suPAR; the $PM_{2.5}$ CI excluded effects of 2% or more per 5 $\mu g\,m^{-3}$. With two prior null cohorts, the authors conclude suPAR is not a meaningful mediator of the PM-CVD link. Limitation: single cross-sectional suPAR measure, and the US contrast of the early 2000s is narrow relative to high-exposure settings, so a null at 5 $\mu g\,m^{-3}$ does not transfer to them. Relevance: low-moderate - a well-powered null worth citing against biomarker-mediation claims; the exposure model, not sensors, sets the contrast.

doi: 10.1097/ee9.0000000000000532

UKB women deprivation $PM_{2.5}$-NO_2 fatty liver 2026 Front Endocrinol (Lausanne)

Socioeconomic disadvantage, air pollution, and metabolic health among women: analyses of UK Biobank and US data

231,525 UK Biobank women: Townsend deprivation predicted FLI (beta 1.114, 95% CI 1.069-1.159) and FLI>=60 (OR 1.080, 1.076-1.084). Age-adjusted pollution associations vanished once deprivation entered the model (FLI beta -0.068, -0.137 to 0.001; FLI>=60 OR 0.999, 0.993-1.004). Across 46 US jurisdictions, +1 SD poverty meant +1.38 points female obesity after $PM_{2.5}$ adjustment. Limitation: adjusting for an area deprivation index that partly lies on the pollution pathway can over-adjust; a 2010 LUR score and state-level US $PM_{2.5}$ are coarse. The null is contextual, not causal. Relevance: low-moderate - shows how easily SES absorbs smooth LUR contrasts; finer exposure would be needed to separate them.

doi: 10.3389/fendo.2026.1948066

Respiratory & allergic — 2 records

Yoon pollutant asthma immune mechanisms review 2026 Allergy Asthma Immunol Res

Impacts and Immune Mechanisms of Pollutant-Mediated Asthma and Airway Inflammation

Narrative review of epithelial injury, barrier disruption, oxidative stress and innate signalling (NF-kB, MAPK, PI3K/Akt, inflammasome) driving T2, non-T2 and mixed asthma phenotypes; pollutant-specific arms include PM-induced NLRP3 activation, O_3 barrier disruption, DEP-driven IL-17A and pollutant-allergen co-exposure. Acknowledges that experimental doses and compositions do not map onto real-world exposure. Limitation: non-systematic, no new data, and no dose metric that a measurement study could target. Relevance: low - background for mechanism sections.

doi: 10.4168/aair.2026.18.5.640

Guangdong COPD surveillance PM_1 $PM_{2.5}$ PM_{10} PAF 2026 Fundamental Research

Association between chronic respiratory diseases and the attributable risk of long-term exposure to atmospheric particulate matters among middle-aged and elderly Chinese population

COPD surveillance sample of 7,371 adults >=40 y (CRD 19.4%, COPD 10.0%). Each 10 $\mu g\,m^{-3}$ of PM was associated with higher prevalence of every CRD outcome, more so in rural residents and biomass users. PAF of CRD: PM_1 19.09% (95% CI 12.41-25.02), $PM_{2.5}$ 18.98% (12.90-24.90), PM_{10} 21.40% (13.84-27.67); COPD highest. Limitation: cross-sectional prevalence, so PAF here is causal only by assumption; $PM_1/PM_{2.5}/PM_{10}$ come from the same satellite-ML product and are near-collinear, so the three PAFs are one estimate three times; biomass fuel confounds rural contrasts. Relevance: low-moderate - PM_1 adds nothing distinct when derived from the same model.

doi: 10.1016/j.fmre.2025.04.013

Reproductive & developmental — 1 record

Guangdong CHD $PM_{2.5}$ constituents 2026 Environ Pollut

Differential Impacts of Fine Particulate Matter Constituents on Congenital Heart Disease across Critical Periconceptional Windows

Guangdong CHD Registry 2004-2022: 12,680 cases, 10,266 controls, multilevel logistic regression by exposure window. Per-IQR increments of $PM_{2.5}$ mass and constituents raised CHD odds 9-16%, strongest in the preconception window and attenuating later; BC and nitrate stayed associated during pregnancy. Conotruncal defects carried the largest ORs (1.14-1.36 per IQR). Mixture models put sulfate and BC first preconception, BC and nitrate thereafter. Limitation: constituents are modelled and mutually collinear, so 'dominant constituent' rankings are mixture-model artefacts as much as biology; registry controls and the higher-SES amplification hint at ascertainment differences. CIs absent from the abstract. Relevance: moderate - constituent-resolved windows need speciation, which optical LCS cannot supply; BC from multi-wavelength nodes is the tractable piece.

doi: 10.1016/j.envpol.2026.129345

Mechanistic toxicology — 1 record

Lung-on-a-chip pollutant toxicity review 2026 Front Bioeng Biotechnol

Microfluidic lung-on-a-chip for evaluating inflammation and toxicity from environmental pollutants

Review of microfluidic lung-on-a-chip platforms - air-liquid interface, cyclic stretch, perfused endothelium - for pollutant toxicology, covering PM, pathogens, VOCs and emerging nanomaterials, with advantages in biomimicry and real-time readout and the obstacles of chip standardisation, operational complexity and subsystem integration. Limitation: no quantitative synthesis; dose delivery (deposited mass per area) is the variable that decides translatability and is not addressed in the abstract. Relevance: low - only connects to monitoring if chips are dosed at measured ambient deposition rates.

doi: 10.3389/fbioe.2026.1942916

Exposure assessment & modelling 7 records

Pu Lima hybrid transfer-learning $PM_{2.5}$ 2026 ACS EST Air

A Hybrid Transfer Learning Approach for Ambient $PM_{2.5}$ Exposure Modeling in LMIC Contexts: A Case Study from Lima, Peru

Lima 2010-2023 daily $PM_{2.5}$ surface from an instance-weighted random forest that fuses reference and low-cost monitor data, satellite AOD and WRF-Chem. Sampling bias toward monitored districts is handled by geo-environmental similarity transfer learning that borrows weighted training instances from the US. Cross-validated R2 0.76; predicted mean $PM_{2.5}$ in mountainous districts fell from 34.19 to 29.23 $\mu g\,m^{-3}$ versus the unweighted model. Limitation: CV R2 is computed on the same biased monitor locations the method claims to correct, so the mountain-region gain is unverified by held-out sites there; how low-cost data were calibrated is not stated. Relevance: high - a template for using sparse LCS networks plus transfer learning in data-poor cities.

doi: 10.1021/acsestair.6c00125

Torii Kalimantan peat fire super-linear scaling 2026 Research Square (preprint)

Peat Burning Scales Super-linearly with Fire-Season Magnitude in the West Kalimantan-Sarawak Border Region, 2012-2025

Fourteen fire seasons of VIIRS detections against a peat map: peat-fire counts scale with total detections as a power law, exponent 1.96 (95% CI 1.57-2.36; R2 0.89), so doubling fire activity quadruples peat burning; robust to alternative predictors (2.36) and a Suomi-NPP-only subset (1.98). Peat share rises from <4% in La Nina years to 30-51% in El Nino years (ONI r 0.78), though ONI is collinear with season size. 24.3% of cells burning >=7 years gave 57.7% of cumulative FRP; scaling survives back-trajectory transport (1.59). Limitation: single-author preprint, detections not emissions, and no PM observations close the chain. Relevance: moderate - receptor-site sensors in Sarawak could test whether haze PM really tracks the squared term.

doi: 10.21203/rs.3.rs-11311408/v1

Monsoon aerosol Pb isotopes rainwater offsets 2026 Atmos Environ

Monsoon-related variability of water-soluble aerosol Pb isotopes and systematic aerosol-rainwater offsets at Peng Chia Yu

METADATA ONLY. No abstract at harvest. Title indicates monsoon-resolved variation in water-soluble aerosol Pb isotope ratios and systematic offsets between aerosol and rainwater signatures. Recorded for coverage; no estimate inferred. Relevance: low - source apportionment input, not a monitoring method.

doi: 10.1016/j.atmosenv.2026.122425

Multi-scale $PM_{2.5}$ nonlinear time-series two metros 2026 Atmos Pollut Res

Deciphering multi-scale $PM_{2.5}$ temporal dynamics and cross-scale interactions in two climatically distinct metropolitan areas: insights from advanced nonlinear time-series analysis

METADATA ONLY. No abstract at harvest. Title indicates nonlinear multi-scale decomposition of $PM_{2.5}$ series and cross-scale interactions in two climatically distinct metropolitan areas (author affiliations suggest Vietnam/Taiwan). Recorded for coverage; no estimate inferred. Relevance: low-moderate - scale separation is how network redundancy and siting density are usually argued.

doi: 10.1016/j.apr.2026.103238

Zhou condensation sink efficiency hygroscopicity 2026 Environ Sci Technol

Condensation Sink Efficiency of Atmospheric Particles Regulated by Their Chemical Compositions and Hygroscopicity

Defines a CS efficiency coefficient (epsilon = CS_eff/CS_calc, the latter assuming irreversible uptake) for gaseous H2SO4. Epsilon depends on composition and water uptake: dry pure organics scavenge less efficiently than inorganics, high RH raises epsilon for both, and mixed particles sit between, set jointly by composition and RH; CS efficiency is lower still for HIO_3. Including epsilon improves cluster-survival and sub-3 nm formation simulations and may explain intense NPF despite high CS in megacities. Limitation: epsilon values and their uncertainty are not given in the abstract, and field transfer is 'exploratory'. Relevance: low-moderate - NPF bursts drive number, not mass, so optical LCS miss them; CPC-equipped nodes would not.

doi: 10.1021/acs.est.6c07189

Harbin ammonium $PM_{2.5}$ and precipitation sources 2026 Environ Pollut

From atmosphere to river: Tracing the origins and deposition of ammonium in both $PM_{2.5}$ and precipitation in Harbin

METADATA ONLY. No abstract at harvest. Title indicates source tracing of ammonium in $PM_{2.5}$ and precipitation in Harbin and its deposition to river systems. Recorded for coverage; no estimate inferred. Relevance: low - secondary inorganic aerosol source apportionment.

doi: 10.1016/j.envpol.2026.129335

Cheng BTH multi-source satellite $PM_{2.5}$ stacking 2026 Atmosphere

Machine Learning-Based Retrieval of $PM_{2.5}$ Concentrations Using Multi-Source Satellite Data in the Beijing-Tianjin-Hebei Region of China

BTH 2019-2021: RF, SVR, XGBoost and linear stacking on four predictor sets with an 80:10:10 split. Test R2 rose from 0.51-0.54 with AOD alone to 0.83-0.84 with all eight predictors; stacking best at R2 0.84, RMSE 12.4, MAE 9.7 $\mu g\,m^{-3}$. Southern/southwestern Hebei highest; winter > summer. Missing AOD limited spatial completeness. Limitation: a random split leaks spatial and temporal autocorrelation, so the test R2 overstates out-of-site skill; no spatial-block CV, and gap filling - the real problem - is left to future work. Relevance: moderate - AOD alone explains half the variance; ground nodes and meteorology carry the rest.

doi: 10.3390/atmos17101010

Sensing, forecasting & instrumentation — 4 records

SEARO air-quality early-warning review 2026 Front Public Health

Prediction accuracy of available early warning systems for air pollution across WHO South-East Asia region: a systematic review

PROSPERO-registered review (to Feb 2026): 14 modelling studies from 839 records, PROBAST+AI risk-of-bias. Hybrid deep-learning forecasters reported the lowest RMSE (0.84-4.61 $\mu g\,m^{-3}$) and R2 up to 0.998 for $PM_{2.5}$; ML reached 98.3% categorical accuracy; CTMs cut bias ~70% only with satellite data assimilation. Most studies were at high risk of bias from absent external validation and overfitting; heterogeneity precluded meta-analysis. Limitation: an R2 of 0.998 for daily $PM_{2.5}$ is a leakage signature, and the review reports such figures without re-analysis. Relevance: moderate - operational forecast skill in the region is unproven; LCS networks could provide the out-of-sample validation these systems lack.

doi: 10.3389/fpubh.2026.1945362

Schafer AERONET lunar AOD assessment 2026 Atmos Meas Tech

An assessment of lunar photometry in AERONET

CIMEL model-T lunar AOD at 492 sites since 2014 with an updated empirical correction to the ROLO lunar irradiance model. At all long-record sites, day-night AOD differences are typically within +/-0.01 at 440 nm and as good or better at longer wavelengths; in 3 h day/night transition windows under low, stable AOD, AOD, Angstrom exponent, water vapour and SDA fine/coarse AOD show statistically negligible differences. Limitation: agreement is shown only under stable low AOD - smoke and dust nights, where night AOD matters most, are untested, and lunar-phase coverage leaves gaps. Relevance: high - validated night-time AOD removes the daytime-only constraint on AOD-$PM_{2.5}$ fusion and gives LCS networks a night-time column reference.

doi: 10.5194/amt-19-6487-2026

Bachour WRF-Chem AOD post-correction hot desert 2026 Atmos Pollut Res

Statistical post-correction of WRF-Chem aerosol optical depth under hot-desert conditions

METADATA ONLY. No abstract at harvest. Title indicates statistical bias correction of WRF-Chem AOD in a hot-desert setting (author group based in Qatar). Recorded for coverage; no estimate inferred. Relevance: moderate - dust-dominated AOD is where both models and optical LCS fail, and corrected CTM fields are common fusion inputs.

doi: 10.1016/j.apr.2026.103233

Epple DLR tire-wear particle chassis dynamometer 2026 Atmosphere

Temperature and Inflation Pressure Effects on Airborne Tire Wear Particle Measurements on a Chassis Dynamometer

Two-level factorial (inflation 1.8/3.8 bar x chamber 11/35 C) on the DLR TRWP system, in two blocks of toe setting. Higher temperature raised cumulative MCPC response 3.6-fold (95% CI 1.8-7.2); higher pressure cut OPS number 18% (12-24%); OPS mass proxy fell ~17% with temperature. Toe >48 arcmin produced rising signals and irregular wear. Reference-condition MCPC response doubled over the campaign, so drift monitoring and randomised blocked run order are prerequisites. Limitation: one tire, one rig; the OPS mass proxy assumes density and refractive index; the ultrafine mode may be volatile, not wear. Relevance: moderate - non-exhaust PM metrology with explicit instrument drift; the same discipline applies to LCS.

doi: 10.3390/atmos17101011

Occupational & indoor — 2 records

Creosote railroad switch PAH workers 2026 Ann Work Expo Health

Assessing PAH exposure during construction work of railroad switches using creosote-impregnated ties

Workers assembling railroad switches from creosote ties: breathing-zone sum of 16 PAHs 120,000-280,000 ng/m3 on a single sampled shift; skin pyrene 0.31-140 ng/cm2, highest on palms and calves; urinary 1-HP rose across the week (1.9-17 umol/mol creatinine), most above the UK guidance value by day 4. A PBTK model attributed uptake mainly to the dermal route, and pyrene under protective clothing implies fabric penetration. Limitation: one air-sampling shift, small crew, and the particle/vapour split is not reported, so the inhalation share rests on the model. Relevance: low for ambient sensing - a reminder that biomarker-based particulate PAH exposure is not reducible to air concentration.

doi: 10.1093/annweh/wxag088

Walter neighbourhood air exchange inhaled dose children 2026 J Expo Sci Environ Epidemiol

Neighborhood-scale disparities in air exchange rates and inhaled pollution dose among school-aged children

METADATA ONLY. No abstract at harvest. Title indicates neighbourhood-scale modelling of housing air exchange rates and the resulting inhaled pollutant dose among school-aged children, framed as a disparity analysis (Northeastern University group). Recorded for coverage; no estimate inferred. Relevance: high if infiltration is parameterised from data - indoor/outdoor LCS pairs measure exactly this ratio.

doi: 10.1038/s41370-026-00989-x

Burden, policy & mitigation 5 records

Gujral Dominici Memphis datacenter $PM_{2.5}$ 2026 Preprints.org (preprint)

Behind-the-Meter Versus Grid Electricity for AI Datacenters: Magnitude, Geography, and Equity of $PM_{2.5}$-Attributable Mortality in Memphis

Two datacenters 12 km apart. At a common 240 MW, grid-supplied C1 carried 10.83 attributable deaths/y (95% interval 7.83-13.84) vs 3.67 (2.65-4.68) for behind-the-meter C2; at real/permitted capacities the order reversed, 11.35 (6.48-17.29) for C2 vs 8.10 (4.60-12.38) for C1. 94% of C1 deaths fell >250 km away; 82% of C2 deaths within 50 km, concentrated in the low-income, high people-of-colour stratum. Limitation: preprint; permitted capacity is not operating capacity, and on-site turbine emission factors and the reduced-form dispersion model are the whole result. Relevance: high - a near-source burden that a local LCS network around the site could verify directly.

doi: 10.20944/preprints202610.0819.v1

GBD 2023 G20 stroke air pollution temperature 2026 Thromb Haemost

Estimates, temporal evolution, and contributors of the stroke burden due to air pollution and non-optimal temperature across G20 nations, 1990-2023

GBD 2023, G20 nations. Age-standardised stroke mortality attributable to air pollution plus non-optimal temperature fell from 52.81 to 17.46/100,000 (1990-2023; EAPC -3.44%, 95% CI -3.57 to -3.30); DALY rate EAPC -3.29%. The decline is driven by household air pollution; ambient PM and heat burdens rose or stalled, highest in low/middle-SDI members, and ARIMA projections to 2040 keep them flat. Limitation: GBD re-tabulation with joint attribution of two risk classes; the ARIMA projection adds no exposure information. Relevance: low - confirms that ambient PM, not household fuel, now carries the residual stroke burden.

doi: 10.1055/a-2977-5555

Atlante Aria e Salute Delphi co-creation 2026 Epidemiol Prev

Co-creation as a decision-making tool in the PNC 'Atlante Aria e Salute' project to define the contents of an integrated atlas

Two Delphi rounds (16 then 22 experts and regional stakeholders) plus plenaries to fix the Italian national air-and-health atlas: data sources (ISTAT cause-specific mortality, hospital discharges), territorial disaggregation, pollutant-outcome pairs graded on causal certainty, public-health relevance and routine-data accuracy, attributable-fraction methods and policy counterfactuals. Limitation: process paper; the abstract is truncated before the final outcome list, and no exposure model is specified. Relevance: low-moderate - a national open-data atlas is a natural consumer of fused monitor/LCS exposure surfaces if its resolution is fine enough.

doi: 10.19191/ep26.4-5.s1.a952.090

Air-health web atlases review Italy 2026 Epidemiol Prev

PNC project 'Atlante Aria e Salute': exploratory web review and quality assessment of open-access atlases on the health effects of air pollution

Protocol-driven review of public air-and-health atlases (Sep-Dec 2024): 1,000 websites screened, 15 potentially relevant, 8 included (3 governmental, 3 academic, 2 foundations). All report $PM_{2.5}$ and/or PM_{10}; NO_2 and O_3 are less common; all but one use annual averages, six give dose-response estimates (mostly all-cause long-term mortality), only two cause-specific. Assessed for interactivity, Flesch readability, WCAG accessibility and transparency. Limitation: eight sites is a thin sample and search-engine screening is not reproducible. Relevance: moderate - public health-impact maps ignore sub-annual and sub-city variation, exactly what dense networks measure.

doi: 10.19191/ep26.4-5.s1.a1026.089

Shandong integrated air-pollutant GHG inventory 2026 Atmosphere

Strategies for Co-Control of Air Pollutants and Greenhouse Gases: Insights from an Integrated Emission Inventory in Shandong Province

2024 integrated inventory for 16 Shandong prefectures, 10 air pollutants and 4 GHGs over 8 source categories. Power/heat and industry gave 86.84% of CO_2; industry dominated SO_2, CO and VOCs; mobile sources NOx; agriculture NH_3, CH4 and N2O; fugitive dust dominated primary PM. Interspecies correlations show partial synergy, so co-control cannot be assumed from shared sources. Limitation: inventory only, no uncertainty bounds in the abstract, and fugitive-dust PM is the least constrained category in any inventory. Relevance: low-moderate - fugitive dust dominance is a coarse-mode signal that $PM_{10}/PM_{2.5}$ ratios from LCS can check locally.

doi: 10.3390/atmos17101016

Corpus register

First author	Focus	Journal	Design	Metric	Region
Cardiovascular & metabolic					
MESA long-term air pollution suPAR null	Long-term ambient air pollution is not associated with circulating soluble urokinase plasminogen activator receptor	*Environ Epidemiol*	Observational - cross-sectional	MESA Air spatiotemporal $PM_{2.5}$, NO2, NOx, O3, light-absorbing carbon (long-term residential)	United States (six MESA metros)
UKB arrhythmia NMR metabolome mediation	Large-scale plasma metabolic profiling identifies signatures and biomarkers associated with air pollution and arrhythmias: A prospective cohort study from UK Biobank	*Ecotoxicol Environ Saf*	Prospective cohort	LUR $PM_{2.5}$, PM_{10}, NO2, NOx at baseline residence	United Kingdom (UK Biobank)
UKB women deprivation $PM_{2.5}$-NO2 fatty liver	Socioeconomic disadvantage, air pollution, and metabolic health among women: analyses of UK Biobank and US data	*Front Endocrinol (Lausanne)*	Observational - cross-sectional	2010 residential $PM_{2.5}$-NO2 score (UK); state $PM_{2.5}$ (US)	United Kingdom (UK Biobank) / United States
Reproductive & developmental					
Guangdong CHD $PM_{2.5}$ constituents	Differential Impacts of Fine Particulate Matter Constituents on Congenital Heart Disease across Critical Periconceptional Windows	*Environ Pollut*	Case-control study	Residential $PM_{2.5}$ mass and constituents (sulfate, nitrate, ammonium, BC, OM) by periconceptional window	China (Guangdong)
Respiratory & allergic					
Guangdong COPD surveillance PM1 $PM_{2.5}$ PM_{10} PAF	Association between chronic respiratory diseases and the attributable risk of long-term exposure to atmospheric particulate matters among middle-aged and elderly Chinese population	*Fundamental Research*	Observational - cross-sectional	722-day mean PM1, $PM_{2.5}$, PM_{10} at 1 km (residential address, 2013-2019)	China (Guangdong)
Yoon pollutant asthma immune mechanisms review	Impacts and Immune Mechanisms of Pollutant-Mediated Asthma and Airway Inflammation	*Allergy Asthma Immunol Res*	Narrative review	PM, DEP, O3, NO2, SO2, VOCs, bioaerosols	Not stated
Mechanistic toxicology					
Lung-on-a-chip pollutant toxicity review	Microfluidic lung-on-a-chip for evaluating inflammation and toxicity from environmental pollutants	*Front Bioeng Biotechnol*	Narrative review	Atmospheric PM, pathogens, VOCs, aromatic hydrocarbons, micro/nanomaterials (in vitro)	Laboratory
Occupational & indoor					
Creosote railroad switch PAH workers	Assessing PAH exposure during construction work of railroad switches using creosote-impregnated ties	*Ann Work Expo Health*	Field measurement	Full-shift breathing-zone PAH (16 EPA priority; particle + vapour), dermal tape strips, urinary 1-HP	Sweden
Walter neighbourhood air exchange inhaled dose children	Neighborhood-scale disparities in air exchange rates and inhaled pollution dose among school-aged children	*J Expo Sci Environ Epidemiol*	Metadata only (no abstract)	Residential air exchange rate and infiltrated PM / inhaled dose	United States
Sensing, forecasting & instrumentation					
Bachour WRF-Chem AOD post-correction hot desert	Statistical post-correction of WRF-Chem aerosol optical depth under hot-desert conditions	*Atmos Pollut Res*	Metadata only (no abstract)	WRF-Chem AOD vs ground photometry	Not stated
Epple DLR tire-wear particle chassis dynamometer	Temperature and Inflation Pressure Effects on Airborne Tire Wear Particle Measurements on a Chassis Dynamometer	*Atmosphere*	Laboratory measurement	Airborne TRWP number (MCPC 7 nm-2 um; OPS 0.3-10 um) and OPS mass proxy	Germany (DLR)
SEARO air-quality early-warning review	Prediction accuracy of available early warning systems for air pollution across WHO South-East Asia region: a systematic review	*Front Public Health*	Systematic review	Forecast $PM_{2.5}$ and other pollutants (RMSE, R2, categorical accuracy, bias)	South-East Asia (India, Bangladesh, Thailand)
Schafer AERONET lunar AOD assessment	An assessment of lunar photometry in AERONET	*Atmos Meas Tech*	Instrument evaluation	Night-time lunar AOD (440-1640 nm), Angstrom exponent, SDA fine/coarse AOD	Global / multi-region (492 AERONET sites)

First author	Focus	Journal	Design	Metric	Region
Exposure assessment & modelling					
Cheng BTH multi-source satellite $PM_{2.5}$ stacking	Machine Learning-Based Retrieval of $PM_{2.5}$ Concentrations Using Multi-Source Satellite Data in the Beijing-Tianjin-Hebei Region of China	*Atmosphere*	Machine-learning model	Satellite AOD + LST, NDVI, tropospheric NO2 + meteorology -> surface $PM_{2.5}$	China (Beijing-Tianjin-Hebei)
Harbin ammonium $PM_{2.5}$ and precipitation sources	From atmosphere to river: Tracing the origins and deposition of ammonium in both $PM_{2.5}$ and precipitation in Harbin	*Environ Pollut*	Metadata only (no abstract)	$PM_{2.5}$ ammonium (isotopic source tracing) and wet deposition	China (Harbin)
Monsoon aerosol Pb isotopes rainwater offsets	Monsoon-related variability of water-soluble aerosol Pb isotopes and systematic aerosol-rainwater offsets at Peng Chia Yu	*Atmos Environ*	Metadata only (no abstract)	Water-soluble aerosol Pb isotopes vs rainwater	Taiwan (Peng Chia Yu)
Multi-scale $PM_{2.5}$ nonlinear time-series two metros	Deciphering multi-scale $PM_{2.5}$ temporal dynamics and cross-scale interactions in two climatically distinct metropolitan areas: insights from advanced nonlinear time-series analysis	*Atmos Pollut Res*	Metadata only (no abstract)	Monitor $PM_{2.5}$ time series (multi-scale decomposition)	Not stated
Pu Lima hybrid transfer-learning $PM_{2.5}$	A Hybrid Transfer Learning Approach for Ambient $PM_{2.5}$ Exposure Modeling in LMIC Contexts: A Case Study from Lima, Peru	*ACS EST Air*	Machine-learning model	Daily 1 km $PM_{2.5}$ from reference + low-cost monitors, MAIAC AOD, WRF-Chem and covariates	Peru (Lima)
Torii Kalimantan peat fire super-linear scaling	Peat Burning Scales Super-linearly with Fire-Season Magnitude in the West Kalimantan-Sarawak Border Region, 2012-2025	*Research Square* (*preprint*)	Satellite time-series analysis	VIIRS active-fire detections on peat vs non-peat; FRP; back-trajectories to Sarawak receptors	Indonesia / Malaysia (Kalimantan-Sarawak)
Zhou condensation sink efficiency hygroscopicity	Condensation Sink Efficiency of Atmospheric Particles Regulated by Their Chemical Compositions and Hygroscopicity	*Environ Sci Technol*	Laboratory + field measurement	Pre-existing particle condensation sink toward H2SO4/HIO3; composition and hygroscopicity	Laboratory
Burden, policy & mitigation					
Air-health web atlases review Italy	PNC project 'Atlante Aria e Salute': exploratory web review and quality assessment of open-access atlases on the health effects of air pollution	*Epidemiol Prev*	Web review	$PM_{2.5}$/PM_{10} (all platforms), NO2 and O3 (fewer)	Global / multi-region
Atlante Aria e Salute Delphi co-creation	Co-creation as a decision-making tool in the PNC 'Atlante Aria e Salute' project to define the contents of an integrated atlas	*Epidemiol Prev*	Delphi consensus	Outdoor $PM_{2.5}$, PM_{10}, NO2, O3 (atlas exposure layers)	Italy
GBD 2023 G20 stroke air pollution temperature	Estimates, temporal evolution, and contributors of the stroke burden due to air pollution and non-optimal temperature across G20 nations, 1990-2023	*Thromb Haemost*	GBD secondary analysis	GBD ambient PM and household solid-fuel pollution; non-optimal temperature	Global / multi-region (G20)
Gujral Dominici Memphis datacenter $PM_{2.5}$	Behind-the-Meter Versus Grid Electricity for AI Datacenters: Magnitude, Geography, and Equity of $PM_{2.5}$-Attributable Mortality in Memphis	*Preprints.org* (*preprint*)	Health impact model	Modelled population-weighted $PM_{2.5}$ from on-site turbines vs grid generation	United States (Memphis)
Shandong integrated air-pollutant GHG inventory	Strategies for Co-Control of Air Pollutants and Greenhouse Gases: Insights from an Integrated Emission Inventory in Shandong Province	*Atmosphere*	Emission inventory	Primary PM, SO2, NOx, VOCs, NH3, CO plus CO2, CH4, N2O by source and city	China (Shandong)

Provenance and method

Entry window **15 October 2026**, single day. Harvester legs: **PubMed** health **6**, sensing **3**; **Europe PMC 46** (0 AGRICOLA retro-index); **Crossref by ISSN 78**; **arXiv** no entry dated 15 October; **OpenAlex** HTTP 429. 129 unique, 22 already carried, 107 fresh. **PubMed connector** (EDAT 15 Oct) 7 PMIDs, 3 appended — **110** candidates screened, **25 admitted**, **85 rejected** with logged reasons. **Consensus** calibration sweep: 10 hits, **0 new** (Ginsburg 2026 and Kumpika 2026 already archived). **Trial watch**: 0 registrations updated 15–16 October. Abstract retry recovered 7 of 12; **5 records carried on metadata alone**. DOI gate 0 fail, 2 advisory warnings (Epidemiologia e Prevenzione DOIs resolve at doi.org but are absent from Crossref). **Scholar Gateway** still `ACCESS_DENIED`. Abstracts remain the copyright of their publishers.